

Crystal-Controlled FM Transmitter



JOY MUKHERJI

Most simple FM transmitters use LC (inductor and capacitor) tuned circuits to generate the carrier waves. But they are inherently unstable due to drift of the resonant frequency. Even moving your body in front of the transmitter changes the frequency or destabilises it altogether. The crystal-controlled FM transmitter presented here is extremely stable. At the heart of the circuit is a phase-locked loop (PLL) clock multiplier chip ICS501.

Circuit and working

Fig. 1 shows the circuit diagram of crystal-controlled FM transmitter. It is built around 5V voltage regulator 7805 (IC1),

low-power audio amplifier LM386 (IC2), a PLL clock multiplier chip ICS501 (IC3), power transistor 2N3866A (T1) and a few other components.

The circuit works on 12V regulated power supply. IC1 provides 5V regulated voltage to drive IC3. The audio from the condenser microphone is amplified in audio amplifier stage built around IC2 and applied to the varactor diode D1 via resistor R3. The audio-frequency signal modulates the bias voltage on the varactor diode, varying the crystal's frequency and thus producing an FM signal at pin 5 of ICS501. This gives us about 40mW at the base of RF power transistor T1. The transistor T1 is biased in class C mode and provides a good RF gain. 50-ohm antenna-matching impedance is formed by inductor L2 and variable capacitors (also called trimmers) VC1 and VC2. Adjust L2, VC1 and VC2 for maximum output power.

The ICS501 uses 12MHz external reference crystal, which is multiplied by a factor of 8 (using S0 S1 as 11). The carrier level is around +27dBm (500mW) at 96MHz (8x12MHz). That is, when you speak into microphone MIC1, you will receive the transmitted

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - LM386 low-power audio amplifier
- IC3 - ICS501 PLL clock multiplier
- D1 - MV209 episcap diode (varactor)
- T1 - 2N3866A npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 10-kilo-ohm
- R2 - 220-ohm
- R3 - 56-kilo-ohm
- R4 - 33-ohm
- R5 - 10-ohm, 0.5W
- R6 - 1-kilo-ohm
- R7 - 220-ohm, 0.5W

Capacitors:

- C1 - 470pF ceramic disk
- C2, C5, C6, C8-C10 - 100nF ceramic disk
- C3, C7 - 220 μ F, 25V electrolytic
- C4 - 10 μ F, 25V electrolytic
- VC1, VC2 - 50pF variable capacitor

Miscellaneous:

- L1 - 12-turn air-core 8mm-dia 26SWG wire
- L2 - 5-turn air-core 8mm-dia 26SWG wire
- MIC1 - Electret condenser microphone
- CON1 - 2-pin terminal connector
- RF C1 - 10 to 20 turns on TV balun core
- S1 - On/off switch
- X₁ - 12MHz crystal oscillator
- ANT. - Antenna around 78cm tall
- Heatsink for T1

Test Points

Test point	Details
TP0	0V
TP1	12V when switch S1 is on
TP2	5V
TP3	Amplified audio signal
TP4	96MHz

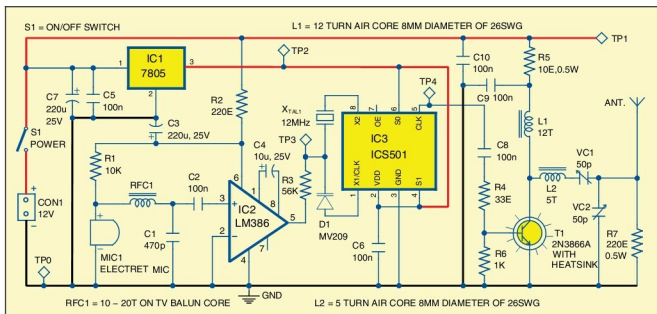


Fig. 1: Circuit diagram of crystal-controlled FM transmitter